CSC1042 Computer Network II

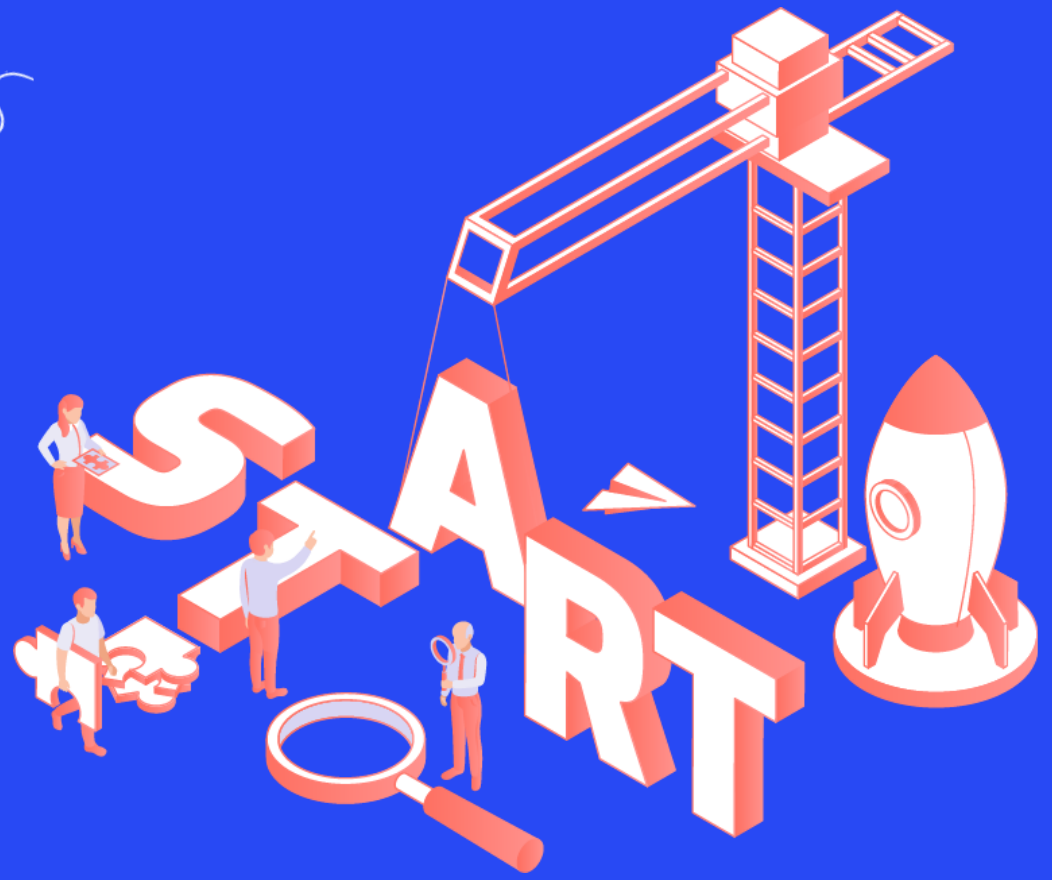
Domain Name System

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Learning outcomes

- Describe the service provided by DNS
- Explain how DNS works
- Describe DNS records

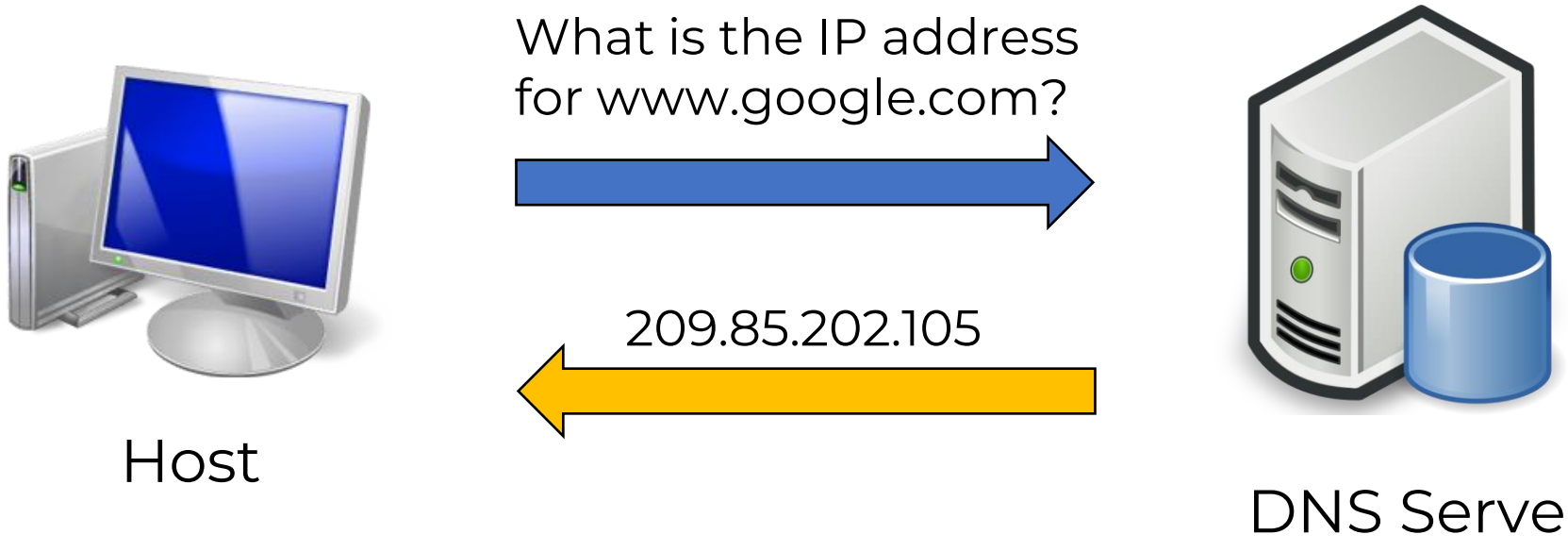


Why we need DNS?

- When you make a connection to a remote server, you usually supply a domain name, e.g., www.google.com
- The TCP/IP stack is used to send/receive data from remote computers
  - This is done using IP addresses
- How does www.google.com get resolved to an IP address?
- Answer: Domain Name System (DNS)

What is DNS?

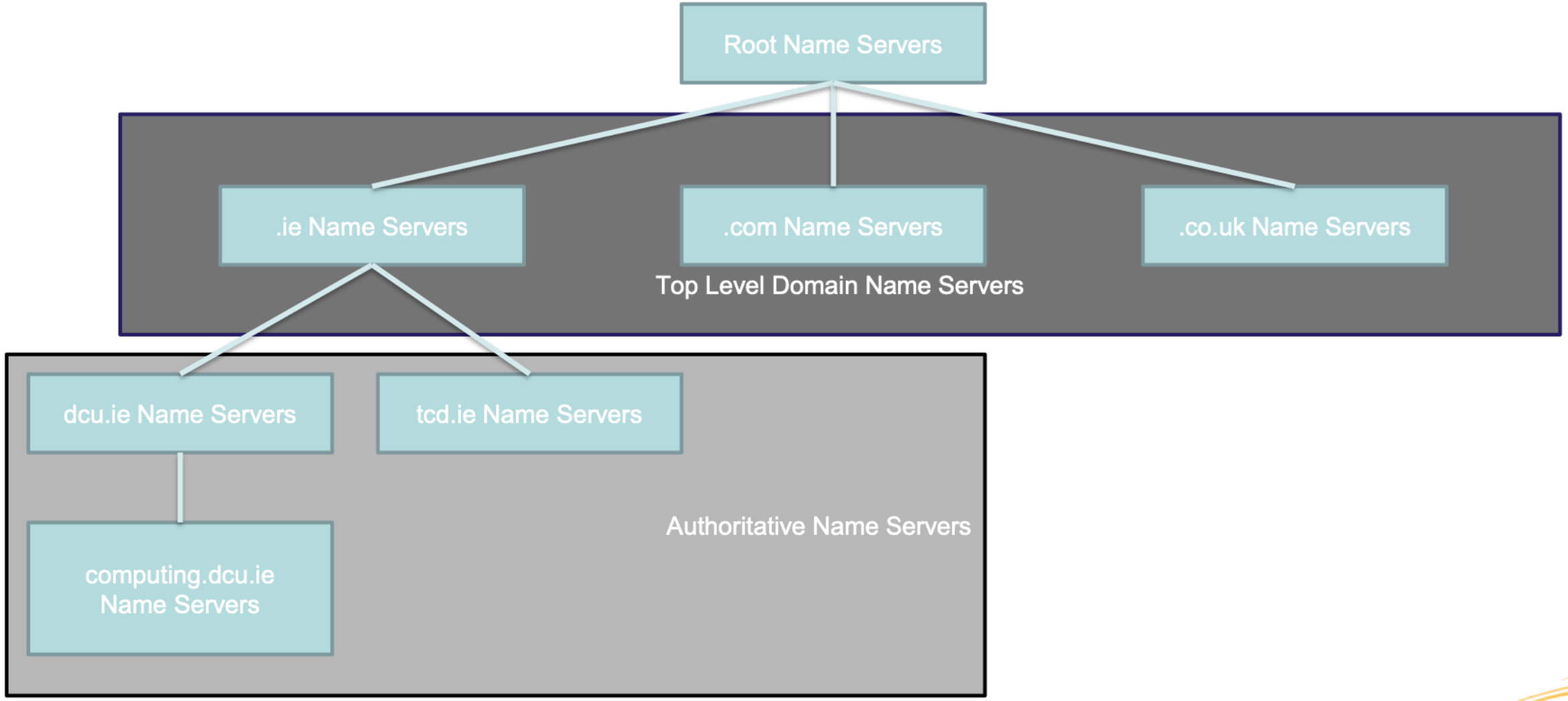
- A server with a database of IP addresses
- When you send it a domain name, it returns an IP address
- This process is called DNS lookup (or DNS resolve)



DNS structure

- The DNS database is *distributed*
  - There is no single DNS database
  - It's split into parts and distributed across the world
- The DNS database is *structured*
  - There are rules guiding what part of the database each server has
- The DNS database is *replicated*
  - Multiple copies of each database exist

DNS structure



Root Name Server

- Contacted first to find out the IP address for a domain
- Decides to which TLD server the request should be forwarded
- There are many root name servers spread across the planet
- Check https://root-servers.org

TLD Name Server

- Manages the DNS mappings for a top-level domain
  - .com
  - .ie
  - .org
- .ie name servers are managed by RIPE NCC
  - https://www.ripe.net

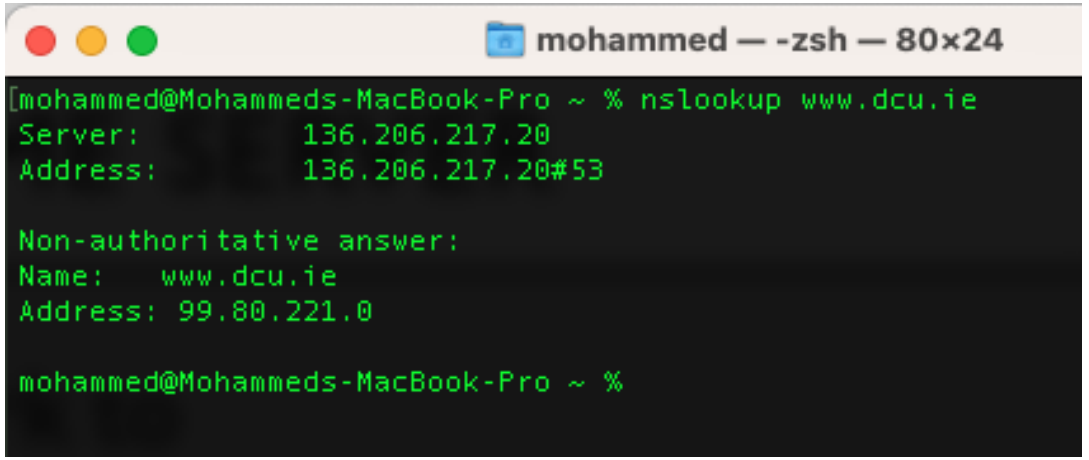
Authoritative Name Server

- DNS mappings provided by an organisation
- e.g., HEANET manages the network infrastructure for third level education institutions in Ireland
- They contain DNS mappings for DCU, TCD, UCD, etc.
- DCU can also be an Authoritative Name Server
  - Mapping for “computing.dcu.ie”, “nursing.dcu.ie”, “physics.duc.ie”
- computing.dcu.ie is also an Authoritative Name Server
  - student.computing.dcu.ie, staff.computing.dcu.ie

Local Name Server

- Usually located on the network to which your computer is attached

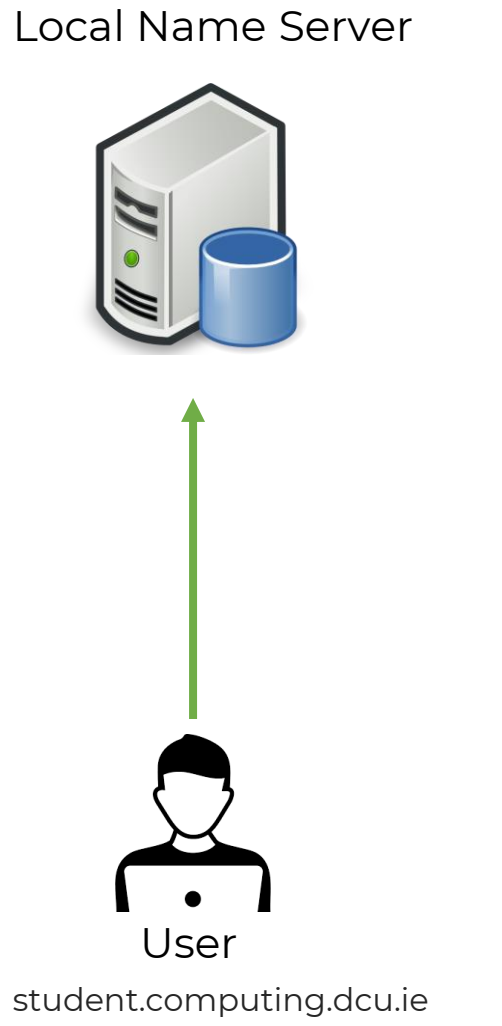
- When you make a request from your home network, you use your network provider's name server
- You can use *nslookup* to determine which name server is being used



- The local name server caches the addresses of common domains
  - e.g., www.google.com

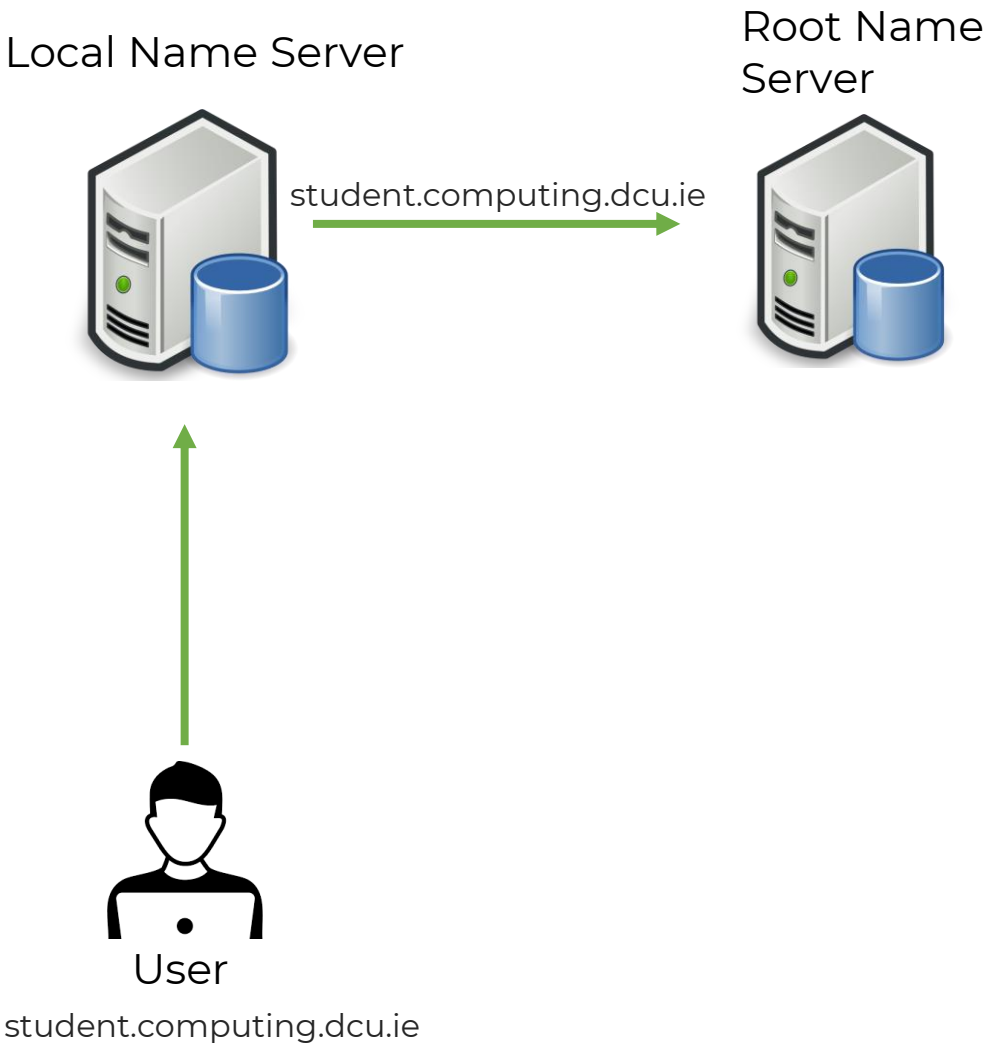
DNS – address resolution

- A user wants to visit student.computing.dcu.ie
- The Local Name Server is first queried



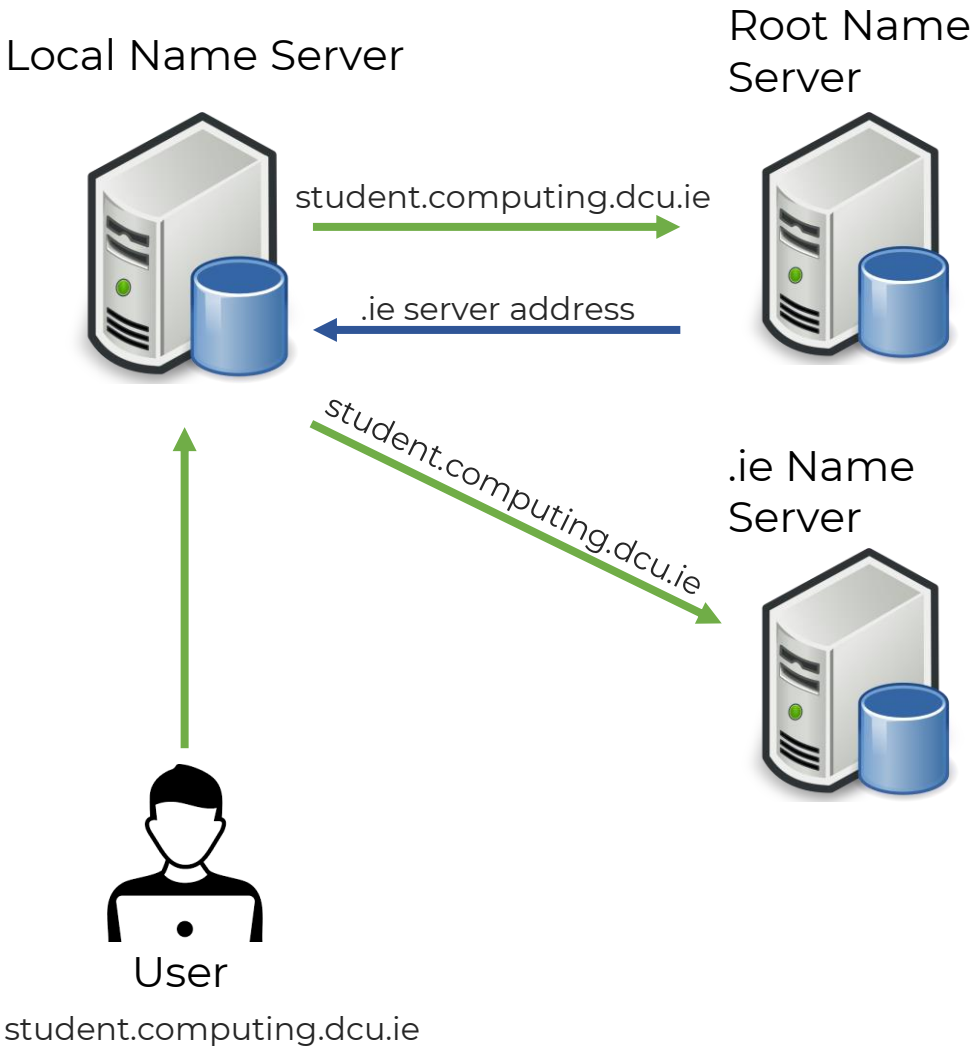
DNS – address resolution

- A user wants to visit student.computing.dcu.ie
- The Local Name Server is first queried
- If it is not in the local name server, the root server is queried



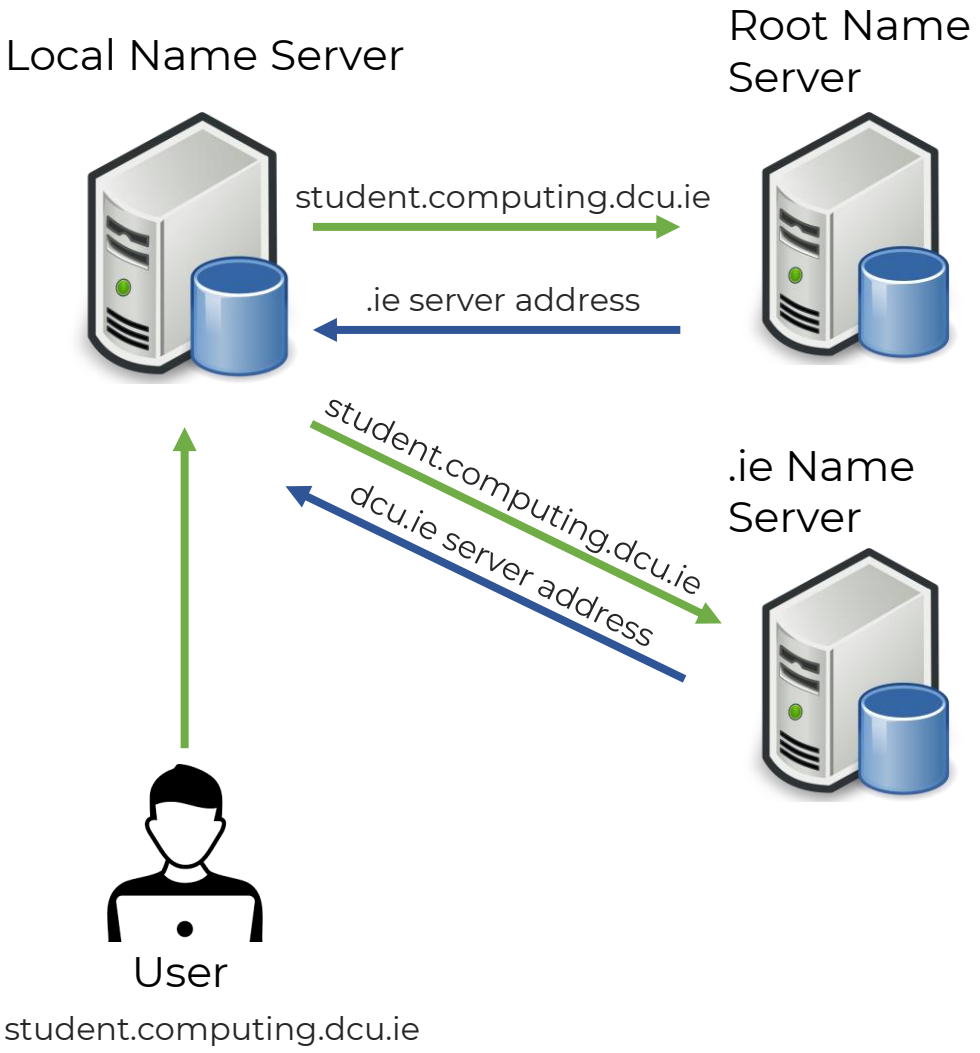
DNS – address resolution

- A user wants to visit student.computing.dcu.ie
- The Local Name Server is first queried
- If it is not in the local name server, the root server is queried
- Root server returns the address of the TLD server
- The .ie server is then queried



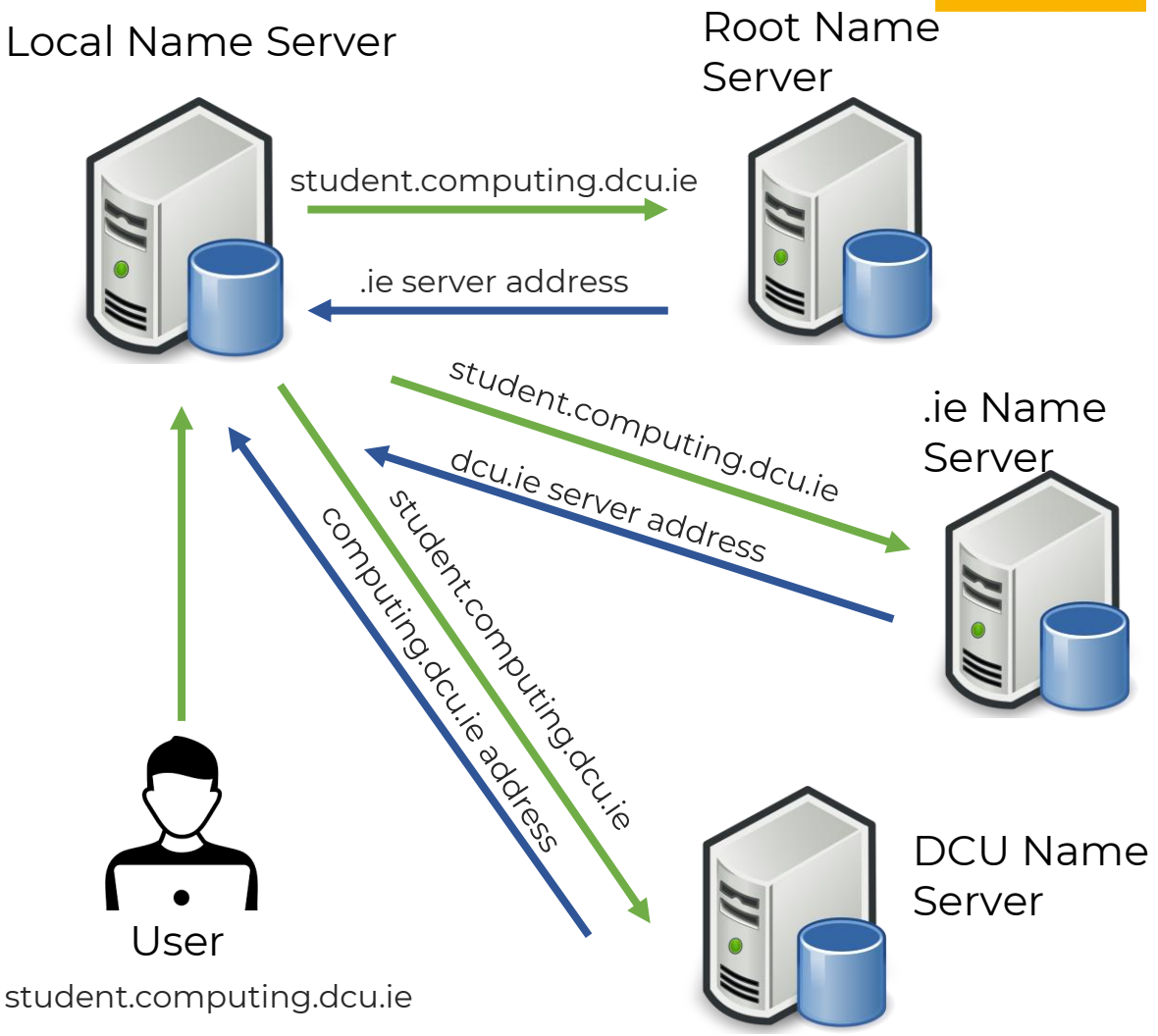
DNS – address resolution

- A user wants to visit student.computing.dcu.ie
- The Local Name Server is first queried
- If it is not in the local name server, the root server is queried
- Root server returns the address of the TLD server
- The .ie server is then queried
- This returns the DCU authoritative name server address



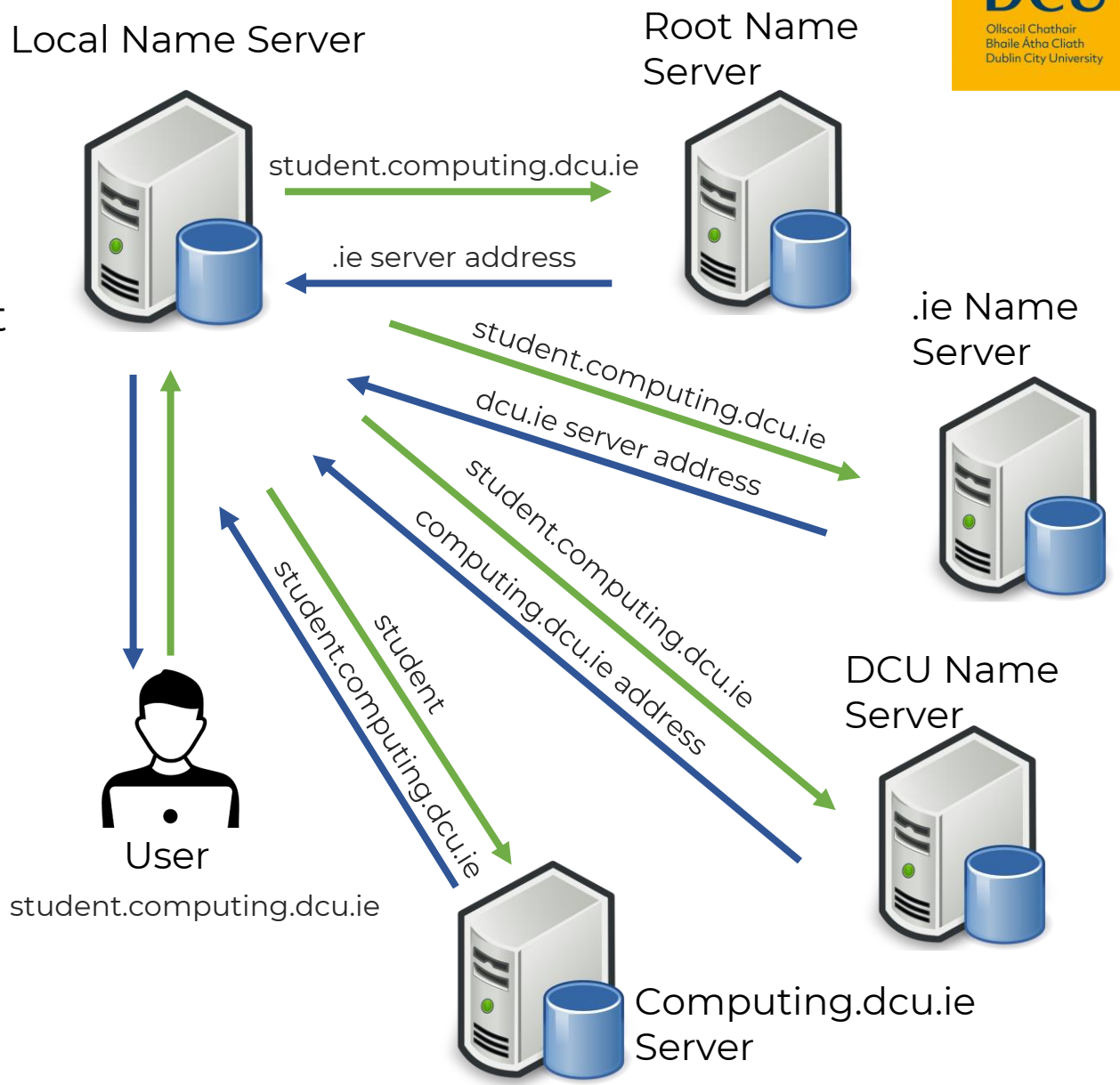
DNS – address resolution

- A user wants to visit student.computing.dcu.ie
- The Local Name Server is first queried
- If it is not in the local name server, the root server is queried
- Root server returns the address of the TLD server
- The .ie server is then queried
- This returns the DCU authoritative name server address
- The DCU name server is then queried which returns the address of computing.dcu.ie address



DNS – address resolution

- A user wants to visit student.computing.dcu.ie
- The Local Name Server is first queried
- If it is not in the local name server, the root server is queried
- Root server returns the address of the TLD server
- The .ie server is then queried
- This returns the DCU authoritative name server address
- The DCU name server is then queried which returns the address of computing.dcu.ie address
- The computing.dcu.ie server is queried which returns the IP address for student.computing.dcu.ie and forwarded to the user



DNS optimisation

- You can see that this process takes time, and DNS servers are queried frequently
- Solution: local name servers store the frequently visited domain names in their cache
  - google.com
  - youtube.com
- They also store domain names based on geography
  - rte.ie
  - met.ie

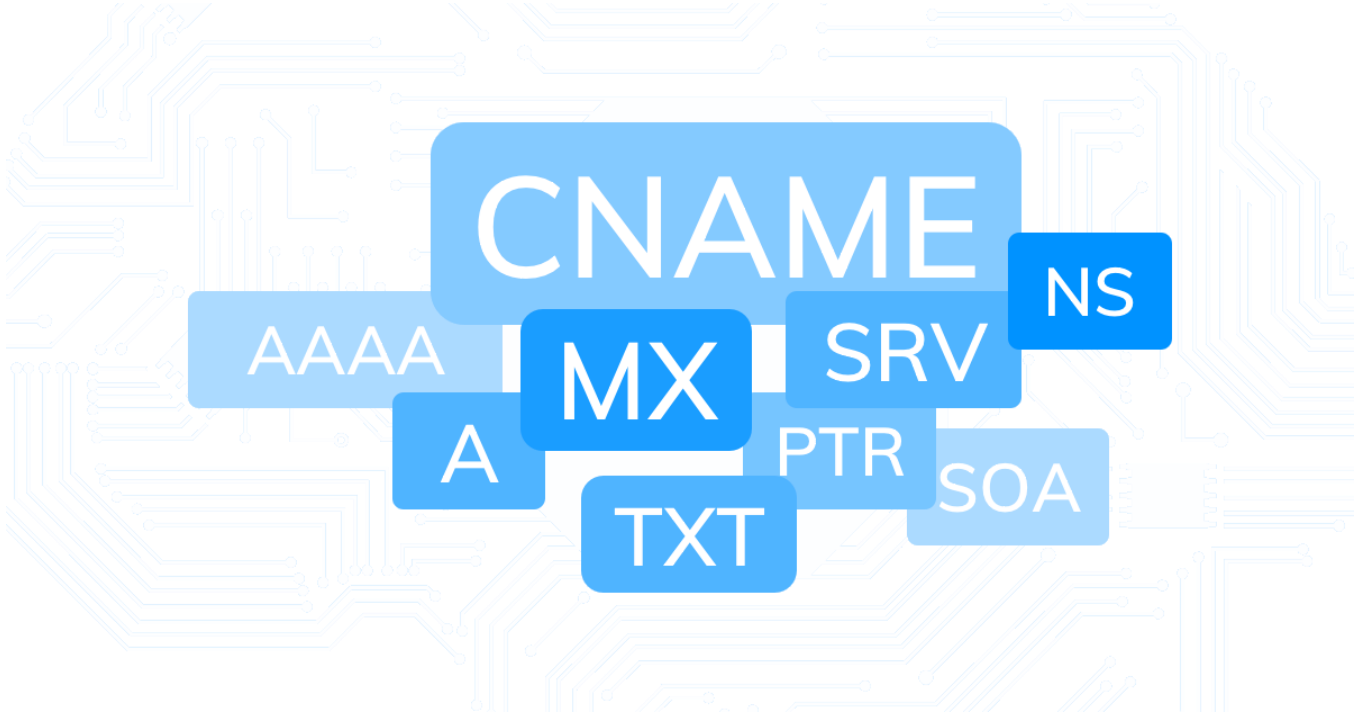
DNS records

- Domain names are stores as DNS records
- If you have a website with a domain name, you need to create DNS records for it
- These are then propagated to the relevant name servers



DNS records

- There are several DNS records
  - Type A
  - CNAME
  - MX
  - NS
  - PTR



Type A

- Used to resolve domain names into IP addresses
- Assume we have the domain name example.com which is assigned the IP address 207.26.21.46
- The type A record is set as follows:

example.com	IN	A	207.26.21.46

- Type AAAA used for IPv6 addresses

example.com	IN	AAAA	2001:db8::ff00:42:8329

Type CNAME

- Used to provide aliases to domain names
- Good when running multiple services from a single IP address
- Assume we run two services: www.example.com and blog.example.com
- The CNAME records are set as follows:



CNAME cannot point directly to an IP address

www	**IN**	**CNAME**	**example.com**
blog	**IN**	**CNAME**	**example.com**

- If the server is queried for blog.example.com, it will return the IP address assigned to example.com

Type MX

- Used to resolve mail servers' addresses
- Assume we have two mail servers: mailhost.example.com and mail.example.com
- The type MX records are set as follows:

example.com	**IN**	**MX**	**10**	**mailhost.example.com**
example.com	**IN**	**MX**	**20**	**mail.example.com**

- If I try to send an email to marie@example.com, it will be resolved to marie@mailhost.example.com
- The numbers 10 and 20 are called MX levels
  - In case no response is received from number 10, try number 20,

Type NS

- Used to list the name servers managing the domain
- It tells the Internet where to find out the domain's IP address
- Assume we have two name servers: ns1 and ns2 that manage the example.com domain
- The type NS records are set as follows:

example.com	**IN**	**NS**	**ns1.example.com**
example.com	**IN**	**NS**	**ns2.example.com**

Type PTR

- Used to provide an address that can resolve the reverse DNS Lookup
  - What is the domain name associated with 207.26.21.46?
  - We should get example.com
- The type PTR record is set as follows:

46	IN	PTR	example.com

Type SOA

- Start of Authority
- Contains administrative info about the DNS zone (server)
- A zone might include the following:
  - One domain name
  - One domain name and several subdomains
  - Multiple domain names
- There can be only one entry per zone

```
; This is the zone file for the domain example.com
$TTL 2d ; This is the default TTL value. It is used in all blank or not defined TTL fields of this file.
$ORIGIN example.com. ; This is the name of the domain for which this zone file has been created.

; The SOA (Start of Authority) record
@       IN      SOA   ns1.example.com.    hostmaster.example.com.    (
                                                2023013100 ; serial-number
                                                12h ; refresh-time
                                                15m ; retry-time
                                                3w ; expiry-time
                                                2h ;  negative-cache-TTL
                                                )

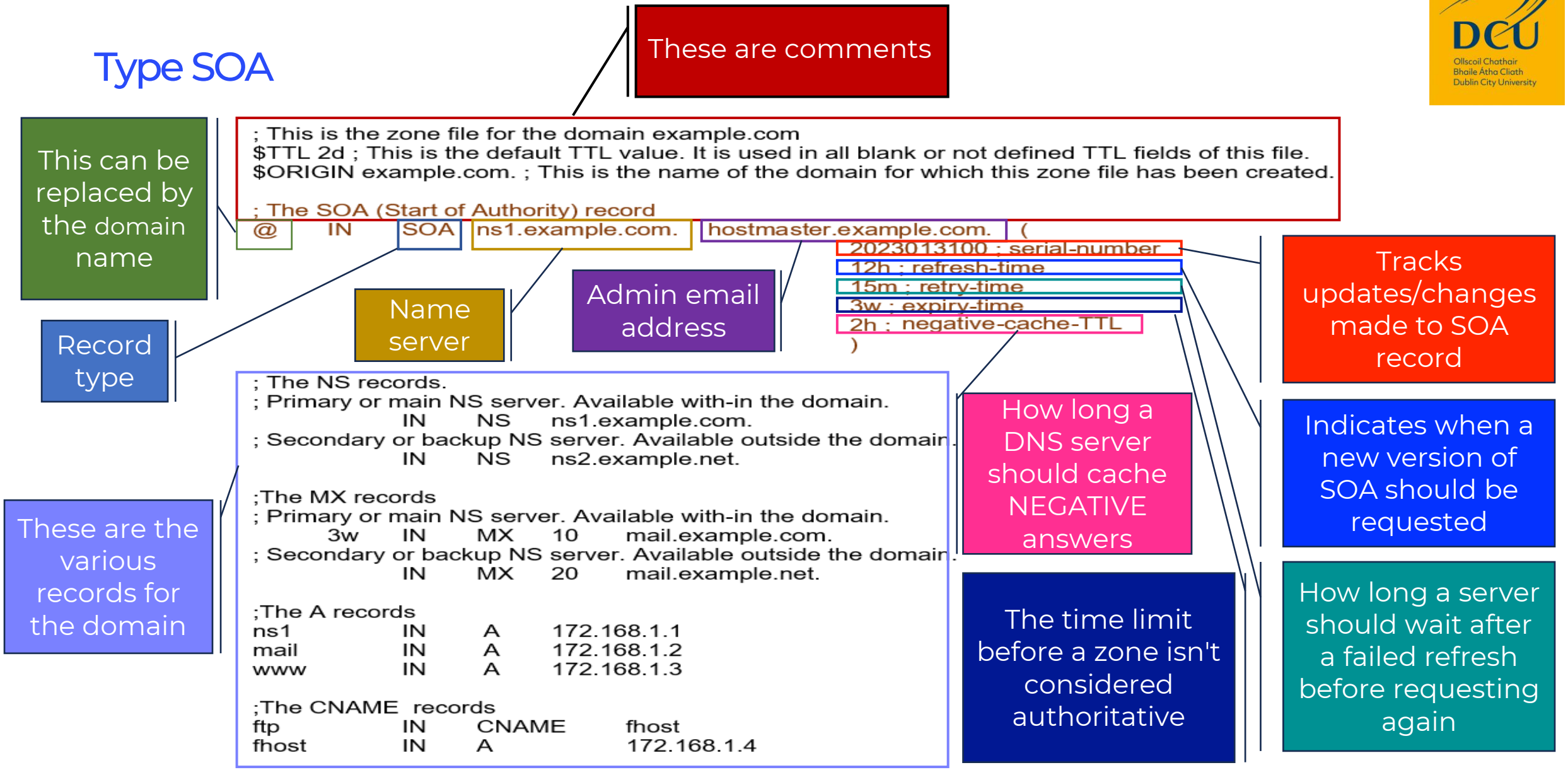
; The NS records.
; Primary or main NS server. Available with-in the domain.
                IN      NS      ns1.example.com.
; Secondary or backup NS server. Available outside the domain.
                IN      NS      ns2.example.net.

;The MX records
; Primary or main NS server. Available with-in the domain.
        3w      IN      MX      10      mail.example.com.
; Secondary or backup NS server. Available outside the domain.
                IN      MX      20      mail.example.net.

;The A records
ns1             IN      A       172.168.1.1
mail            IN      A       172.168.1.2
www             IN      A       172.168.1.3

;The CNAME  records
ftp             IN      CNAME   fhost
fhost           IN      A               172.168.1.4
```

Type SOA


These are comments
This can be replaced by the domain name
; This is the zone file for the domain example.com
$TTL 2d ; This is the default TTL value. It is used in all blank or not defined TTL fields of this file.
$ORIGIN example.com. ; This is the name of the domain for which this zone file has been created.
; The SOA (Start of Authority) record
@ IN SOA ns1.example.com. hostmaster.example.com. (
2023013100 ; serial-number
12h ; refresh-time
15m ; retry-time
3w ; expiry-time
2h ; negative-cache-TTL
)
Record type
Name server
Admin email address
Tracks updates/changes made to SOA record
Indicates when a new version of SOA should be requested
How long a server should wait after a failed refresh before requesting again
How long a DNS server should cache NEGATIVE answers
The time limit before a zone isn't considered authoritative
These are the various records for the domain
; The NS records.
; Primary or main NS server. Available with-in the domain.
IN NS ns1.example.com.
; Secondary or backup NS server. Available outside the domain.
IN NS ns2.example.net.
;The MX records
; Primary or main NS server. Available with-in the domain.
3w IN MX 10 mail.example.com.
; Secondary or backup NS server. Available outside the domain.
IN MX 20 mail.example.net.
;The A records
ns1 IN A 172.168.1.1
mail IN A 172.168.1.2
www IN A 172.168.1.3
;The CNAME records
ftp IN CNAME fhost
fhost IN A 172.168.1.4

THANK YOU